Supplementary Information

Time-domain noise as an evaluation axis for raw-signal nanopore mapping

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Three observations made during this work are reported here rather than in the main text, because each rests in part on a measurement for which no run output was captured. Those values are marked with a dagger (†). None of them is used in any claim in the main text, and `scripts/verify_dwell_journal_numbers.py` lists each as outstanding, together with the run that would settle it. Section and table numbers without an S refer to the main text.

S1 Live accept rates against Icarust

With the same client, reference and run length, the fixed-width arm accepted 43.5%† of reads and the detected arm 9.6%† against stock Icarust, and 0.5%† and 9.6%† against the patched simulator at CV 0.35. Taken at face value, stock Icarust reports the fixed-width arm as 4.5-fold† better, and the patched simulator reverses the order, which is the reversal of Section 3.1 observed on the live interface. These are accept rates on the live loop and are not comparable with the offline true-positive rates (TPR) of Table 1. Re-running stock and patched Icarust against the client would settle them.

S2 Calibration target for event detection

The calibrated detector, threshold 3.0, emits 1562 events per on-target read at CV 0.44, against an ideal of approximately 2000, one event per base, which is 21.9% under-segmentation; seed 42 gives 14.2% TPR in the captured log. In a development run that was not captured, threshold 2.0 gave 2017† events per read, close to the ideal, and 2.0%† TPR, 7.1-fold† lower. An argument offered during development attributed the difference to frame displacement: a spurious boundary shifts every later key in the read. That argument does not distinguish the two error types, because a missed boundary removes an event and shifts every later event index by one exactly as a spurious boundary adds one. A sufficient explanation is simpler. The event count measures net error, insertions minus deletions, and is blind to errors that cancel, so a detector that emits the ideal number of events may still place many boundaries wrongly. The recommendation that follows does not depend on the uncaptured values: calibrate a detector against agreement or indel fraction (Section 2.6), not against its event count.

S3 Calibration transfer between generators

A detector threshold calibrated against signal from a generator written for this project, applied unchanged to Squigulator signal, gave 0.6%† TPR at CV 0, where the Squigulator-calibrated detector gives 18.4% [17.0–19.8] (Table 1), a 30.7-fold† difference. The two generators differ in their amplitude-noise models, and the detector threshold is a t-statistic whose operating point depends on that model. Whatever the magnitude, a result obtained with event detection is interpretable only together with a statement of how the detector was calibrated and on what signal.